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4/17/25

LAB 10

DC motors

Introduction

The goal of this lab was to understand how a motor works and the relationship between a motor's load, torque, speed, power and efficiency. Firstly, it is important to understand how a motor produces action and force. A motor creates motion by using magnets. Electrical energy is converted into magnetic force and finally into kinetic energy (action). There are two pole motors, and the magnetic field produced by the armature windings is either repelled or attracted by the stators permanent magnets, which produces a rotational movement. So the rotational speed of a DC motor depends on the interaction between the two magnetic fields- one set up by the stators' stationary magnets and the other by the rotating electromagnets. Therefore, by controlling this interaction we can control the speed of rotation. Note that the the stator's permanent magnets are fixed and not changeable but we can alter the strength of the armatures electromagnetic field by controlling the current flowing through the windings, resulting in more or less magnetic flux (measure of total magnetic field which is in a given surface) being produced and thus a stronger/ weaker interaction and a faster/ slower speed. As the armature rotates, electrical current is passed from the motors terminals to the next set of armature windings via carbon brushes

Relevant equations for motors:

- Horsepower = $\frac{torque * speed}{5252} = \frac{work}{time}$

  Torque is in pounds- feet, speed is in rev per minute (RPM) and 5252 is derived from 33,000 (the number of foot-pounds of work a horse can do per minute) by 2pi (a full revolution).

- $T = k\phi I$

  *torque = constant of work * flux *current*

- $Eg = k\phi s$

  *generator voltage = machine constant * flux* speed, recall e= generated voltage*

- $\xi = \frac{pout}{pin} < 100\%$

  *Pout is the mechanical power in HP ($\frac{torque * speed}{5252}$) multiplied by 746, because 1 HP =746 watts. Pin = V *I*

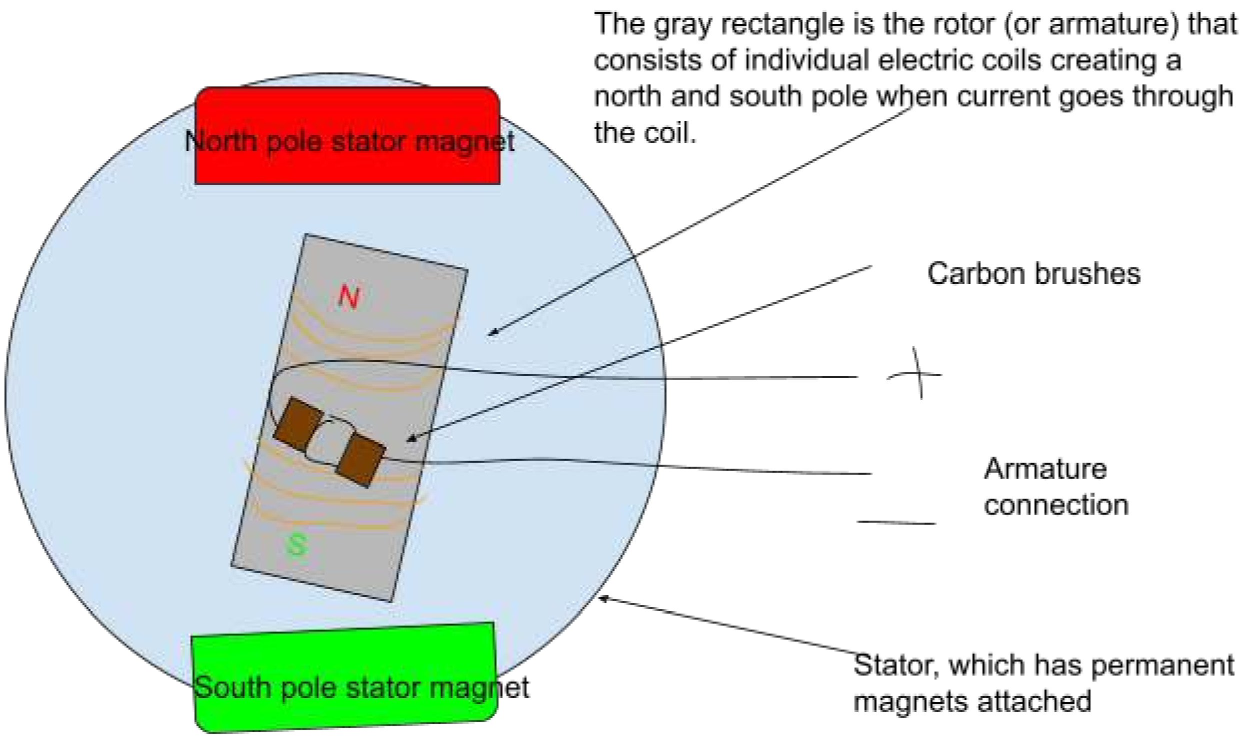
- $E = blv$

  *induced voltage E = magnetic field B *length of coil * speed V*

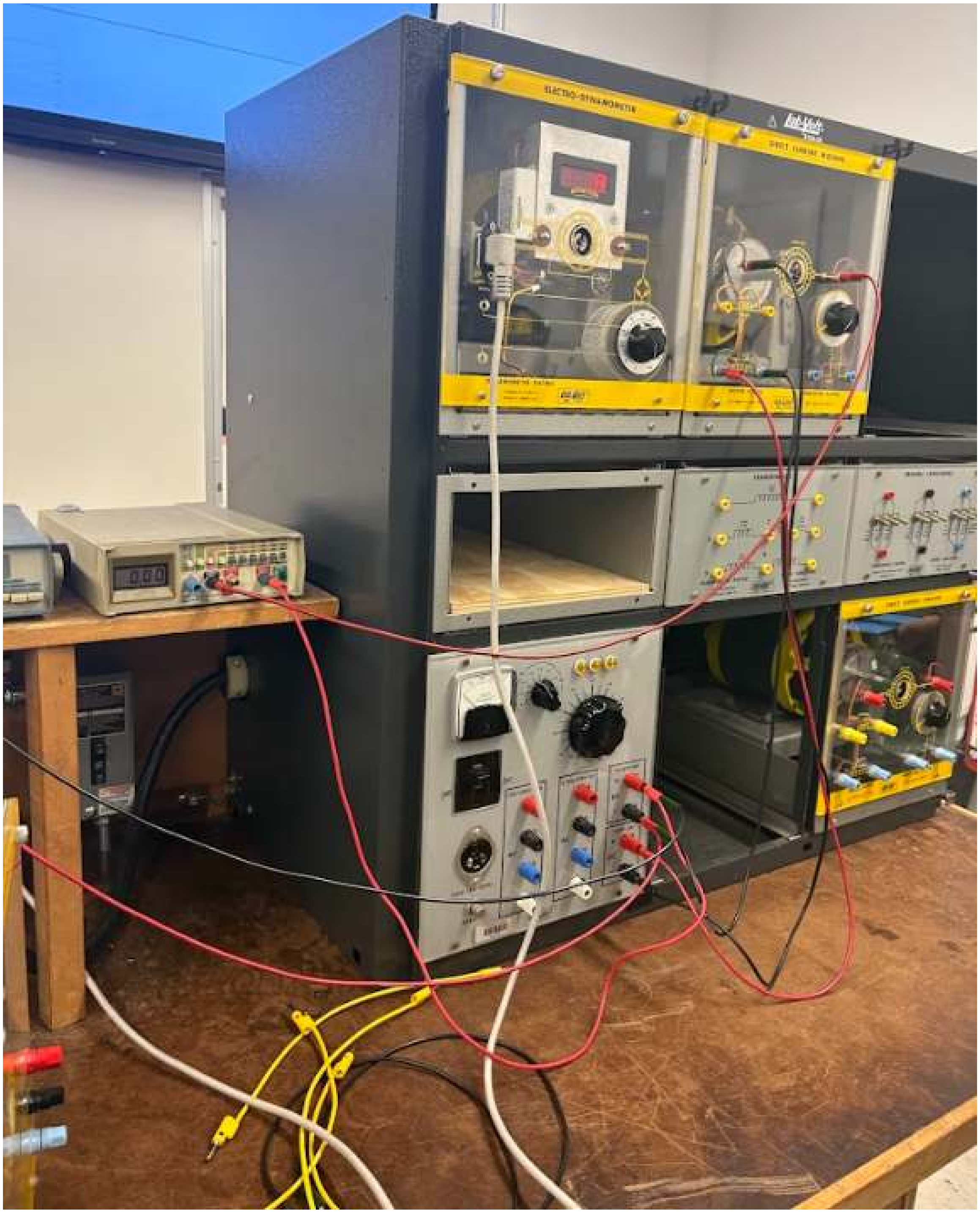
- F=BIL

  *force= magnetic field * current in armature * length of conductor in magnetic field.*

Roles of DC motor components

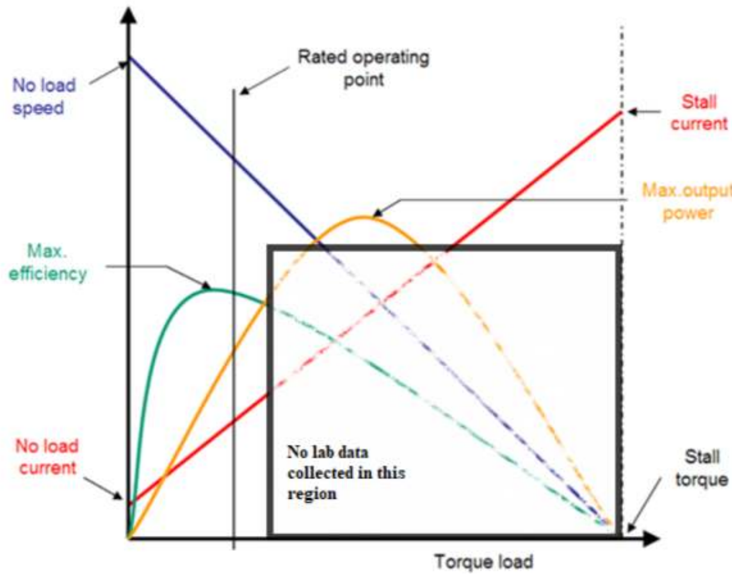


IMG 1. *Roles of DC motor components*

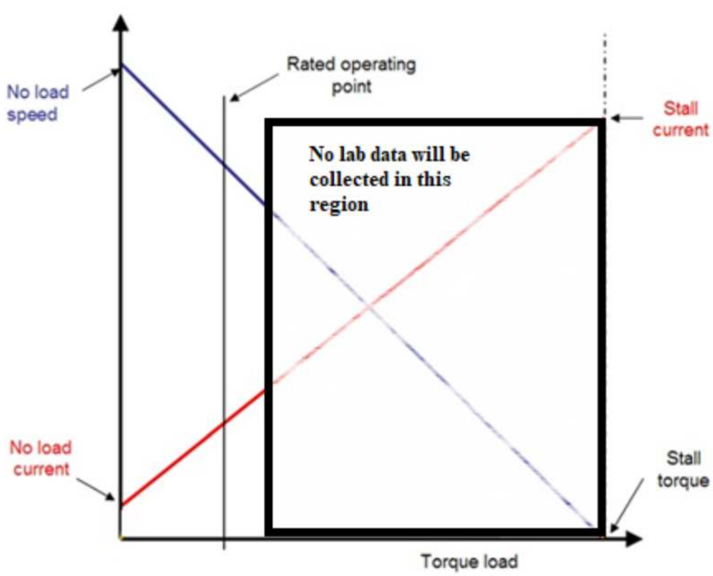


IMG 2. *Physical circuit*

In this lab we dealt with high voltage and did not want to stress the motor. Because of this, we could not gather data to the left of the rated operating point, and making a graph of our data would be not particularly representative of how a DC motor works



IMG 3. *A demonstration of the limitations of our gathered data.*



IMG 4. *A demonstration of the limitations of our gathered data.*

Data

Voltage (volts)	Torque (N-m)	Speed (RPM)	Armature current (Amps)	Load	Pin (watts)	Pout (watts)	Efficiency %
104	15.3	918	2.65	50	1471	275.6	18
105.1	10.2	945	2.29	45	1009.72	240.7	24
112.6	2.2	1177	0.59	40	270.89	66.43	25
109	4.5	1045	1.49	35	491.74	162.41	33
122.8	0.8	1300	0.35	30	108.86	42.98	39
111.7	2.4	1115	1.1	25	280.20	122.87	44
120.9	0.2	1350	0.43	20	51.99	28.27	53
113.6	1	1155	0.94	15	120.99	106.78	88
114.5	0.6	1175	0.86	5	98.47	73.87	74

Table 1. *Data*

As an example, we calculated the first voltage point as so:

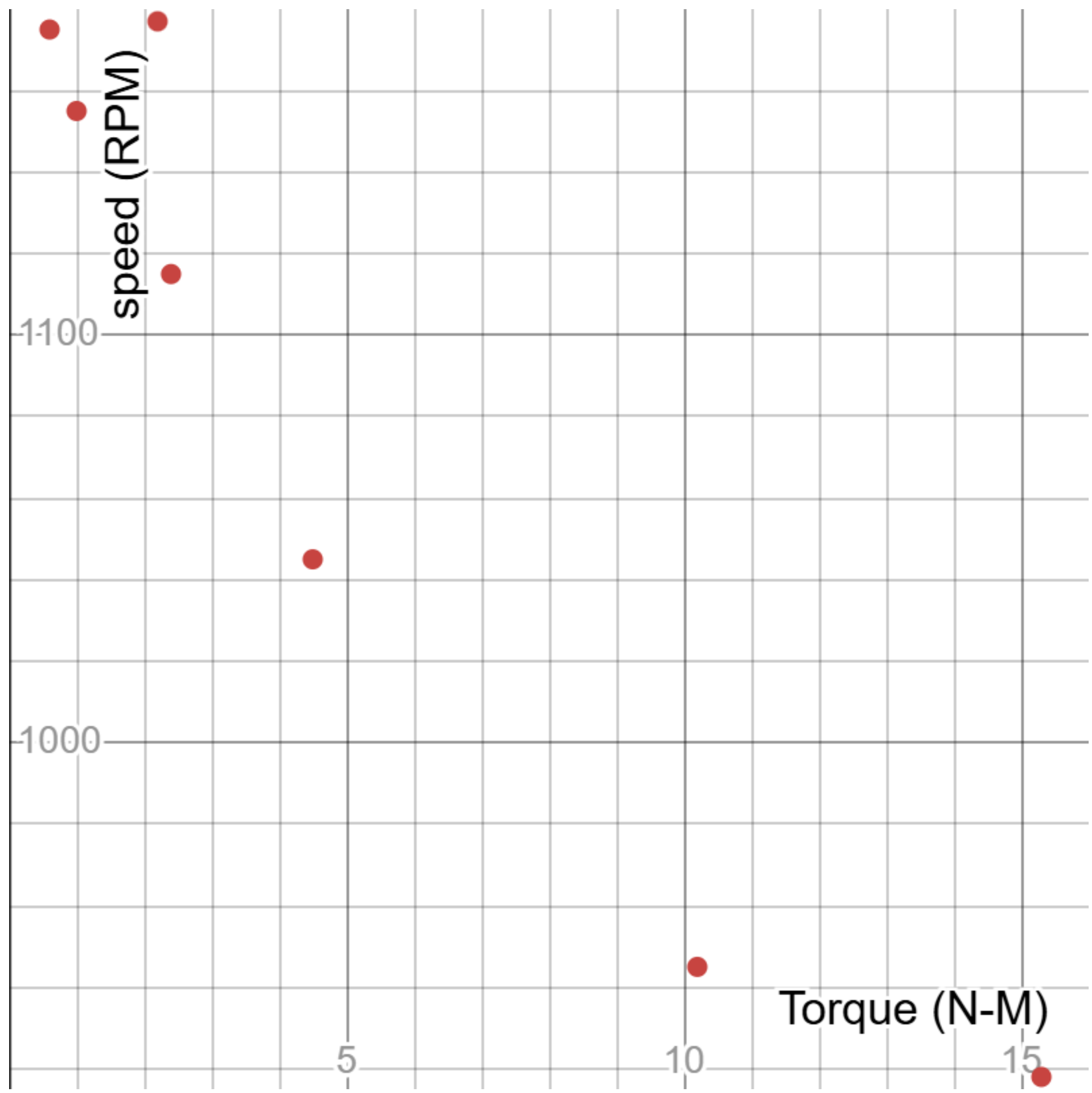
Recall $\xi = \frac{pout}{pin} < 100\%$

To calculate the input power = *Hp=(* $\frac{torque * speed}{5252}$)= $\frac{0.737\ lb-ft * 15.3\ RPM}{5252}$ = 1.972 HP.

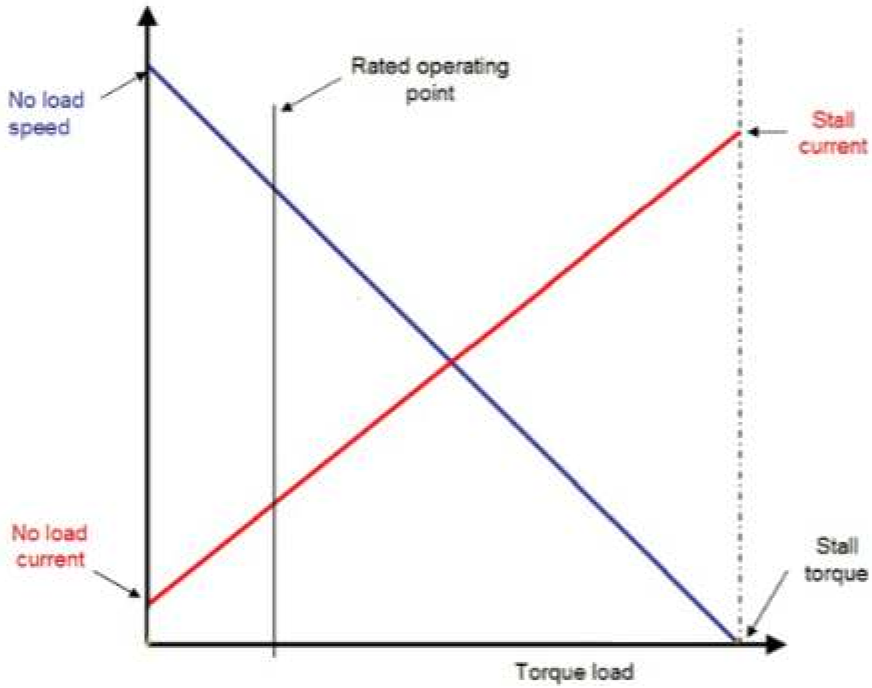
Changing it into watts, we get $1.972\ *\ 746\ watts =\ 1471\ watts$

To calculate the output power, = *V *I = 104 * 2.65 =275.6*

To calculate the efficiency, $\frac{pout}{pin} = \frac{275.6}{1471} = 18\%$



IMG 5. *Torque vs speed. This graph is limited by our capabilities to gather data, so explanations about the meaning of the graph and points can be found in the conclusion*



IMG 6. *Torque vs speed.*

Both plots from images 5 and 6 show that the motor speed is inversely proportional to load torque. This means that the motor slows down as it's loaded . An increasing output/ load torque means more current is ended by the motor and so the input current increases./ This means that the torque is proportional to the current, and there's a torque constant for a lot of motors which is related to the slope of the current line. It can be calculated as torque Kt * I. Kt is the slope of the line if torque versus current is plotted. Also a motor can be linked to a transformer that as the secondary load increases (More Is current) the input or primary current IP must increase to provide for this secondary current, similarly the output load torque and input current are directly related.

Conclusion

We saw in the lab that the speed depends on the load as well as the voltage. Factors like friction and going downhill are similar to increasing the load as we did in the lab, and have the same effects of increasing the torque while decreasing the speed and generated voltage. We also saw that there is clearly a range that the motor works less efficiently after. Generally speaking, we want to operate the motor to the left of the rated operating point (keeping in mind that for DC motors, the X axis is torque and the Y axis is speed) ALthough the rated operating point is not at the peak efficiency, operating to the right of this point can lead to many issues. One issue is that the load can be too great for the motor to run and it will have a stall current. Contrairly, if there is no load (a hypothetical situation) the motor would theoretically be at its highest speed. It would still need a bit of current, though.